MPPT-CONTROLLED GRID-CONNECTED SOLAR PV SYSTEM

DISTRIBUTION AUTOMATION AND SMART GRID (EC9130)

FINAL PROJECT REPORT

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ABBREVIATIONS AND ACRONYMS

AC	:	Alternating Current
CLC	:	Current Loop Control
DC	:	Direct Current
DER	:	Distributed Energy Resources
FFT	:	Fast Fourier Transform
LCL	:	Inductor-Capacitor-Inductor Filter
MPP	:	Maximum Power Point
MPPT	:	Maximum Power Point Tracking
P&O	:	Pertub and Observe
PI	:	Proportional Integrator
PLL	:	Phase- Locked Loop
PV	:	Photovoltaic
RMS	:	Root Mean Square
SVPWM	:	Space Vector Pulse Width Modulation
THD	:	Total Harmonic Distortion

Chapter 1: INTRODUCTION

1.1 Motivation and Overview

The increasing demand for clean and sustainable energy has made solar photovoltaic (PV) systems one of the most important renewable energy technologies in modern power networks. Among the different configurations, grid-connected solar PV systems are especially attractive because they can deliver power directly to the utility grid while reducing dependence on conventional fossil-fuel-based generation.

A practical challenge in PV systems is that the electrical output of the solar array changes continuously with solar irradiance and cell temperature. Because of this variation, the array does not naturally operate at its best power point all the time. If proper control is not used, a significant amount of available solar energy can be lost. This creates the need for a Maximum Power Point Tracking (MPPT) method that can continuously adjust the operating point of the PV array and extract the highest possible power under changing environmental conditions.

In addition to maximum power extraction, a grid-connected PV system must also deliver stable and good-quality power to the utility network. The Direct Current (DC) power generated by the array has to be processed through a boost converter, regulated at the DC-link, converted to Alternating Current (AC) by a three-phase inverter, synchronized with the grid voltage and frequency, and filtered before injection. Any weakness in these stages can lead to poor efficiency, unstable operation, or unacceptable harmonics.

This project therefore focuses on the modeling, control, and evaluation of an MPPT-controlled grid-connected solar PV system using MATLAB/Simulink. The work brings together the PV array, Perturb and Observe (P&O) based MPPT, DC-DC boost converter, DC-link voltage control, Phase-Locked Loop (PLL), current loop control (CLC), inverter, and LCL filter in one integrated simulation platform. The final objective is to create a system that can extract maximum solar power, inject synchronized three-phase current into the grid, and maintain good power quality under varying operating conditions.

1.2 Aims and Objectives

The main aim of this project is to design and evaluate an MPPT-controlled three-phase grid-connected solar PV system using MATLAB/Simulink.

The specific objectives of the project are:

1. To model a solar PV array suitable for grid-connected operation under varying irradiance and temperature conditions.
2. To implement a P&O based MPPT algorithm to extract the maximum available power from the PV array.
3. To design a DC-DC boost converter to regulate the PV output and interface it with the inverter stage.
4. To maintain a stable DC-link voltage through outer-loop PI control.
5. To synchronize the inverter output with the utility grid using a PLL.
6. To implement d-q frame current loop control for accurate active current injection and zero reactive current operation.
7. To reduce output harmonics using an LCL filter and verify power quality through FFT-based THD analysis.
8. To evaluate the overall system response under dynamic irradiance conditions.

Table 1. 1: Summary of main project objectives

Objective Area	Purpose
MPPT control	Extract the highest available power from the PV array.
DC-link regulation	Provide a stable DC source for the inverter.
PLL and current control	Synchronize the inverter with the grid and regulate active/reactive current.
LCL filtering and THD analysis	Improve injected current quality and verify compliance with the project target of THD below 5%.

1.3 Research Scope

This project is limited to the simulation, control, and performance evaluation of a three-phase grid-connected PV system. The complete work is carried out in MATLAB/Simulink, and the study focuses on the interaction between the PV array, MPPT controller, boost converter, DC-link voltage controller, inverter, PLL, current controller, and LCL filter. The scope includes dynamic irradiance testing, current tracking analysis, DC-link voltage regulation, and THD measurement at the grid side.

The present study does not include hardware implementation, real-time embedded control, or experimental validation using a physical prototype. It also does not compare a large number of MPPT algorithms experimentally; instead, it focuses on the design and validation of a P&O-based method within the selected simulation platform. Therefore, the conclusions drawn in this report are limited to simulation-based performance, though they provide a strong foundation for future hardware-oriented work.

Chapter 2: LITERATURE REVIEW

2.1 Introduction

Grid-connected PV systems have become increasingly important because they convert clean solar energy into electrical power that can be directly supplied to the utility grid. A standard grid-connected PV system usually includes a solar array, MPPT controller, DC-DC converter, DC-link capacitor, inverter, synchronization mechanism, and output filter. The effectiveness of the entire system depends not only on how much power can be extracted from the PV array, but also on how well that power can be delivered to the grid with minimum distortion and stable operation. Your uploaded project documents reflect this same system-level view and emphasize the need for proper control at each stage.

A major challenge in PV systems is that the operating point of the array changes with irradiance and temperature. If the PV array is allowed to operate freely, it may not remain at its maximum power point, causing energy loss. For this reason, MPPT techniques are widely used in PV systems to continuously adjust the operating point so that maximum available power can be extracted. Among the different techniques, Perturb and Observe remains one of the most commonly used methods due to its simplicity and ease of implementation. Your project presentation also identifies P&O as the selected tracking method and highlights the need to improve response under changing conditions.

2.2 Review of MPPT and Inverter Control Approaches Performance Metrics

Among the many MPPT techniques available, Perturb and Observe remains one of the most widely used due to its simple logic, ease of implementation, and low computational burden. The basic idea is to slightly perturb the operating point of the PV array and observe the resulting power change. If power increases, the controller continues in the same direction; if power decreases, it reverses the perturbation direction. Although this method is practical, conventional P&O can show oscillations near the maximum power point and slower response under rapidly changing environmental conditions.

To address these weaknesses, improved P&O strategies are often introduced. These may include refined step size selection, boundary checking, better initialization, or more adaptive perturbation behavior. Such enhancements reduce unnecessary oscillation around the MPP and improve convergence speed when irradiance changes suddenly.

For the inverter stage, synchronization with the grid is a critical requirement. A Phase-Locked Loop is commonly used for this purpose. By transforming the measured three-phase grid voltage into a rotating d-q reference frame, the PLL estimates the grid angle and frequency so that the inverter can remain synchronized. Once the rotating frame is available, current control becomes easier because the active and reactive components can be regulated separately by PI controllers.

Current Loop Control in the d-q frame is especially useful in three-phase grid-connected converters. The d-axis current is normally related to active power transfer, while the q-axis current is associated with reactive power. By controlling these two quantities independently, the inverter can deliver the required active power to the grid while keeping reactive power close to zero for unity power factor operation.

Because inverter switching introduces harmonics, output filtering is also a standard requirement. LCL filters are commonly selected in grid-connected systems because they provide stronger attenuation of high-frequency harmonics compared with simple L or LC filters. As a result, they help improve the quality of the current injected into the utility network.

2.3 Performance Metrics

The performance of a grid-connected solar PV system can be assessed using several key technical indicators. One of the most important is MPPT effectiveness, which describes how quickly and accurately the controller tracks the maximum power point under changing irradiance or temperature. Another major indicator is DC-link voltage stability, because the inverter depends on a properly regulated DC bus for consistent AC generation.

Synchronization quality is another important metric. Proper synchronization is reflected by balanced three-phase voltage and current waveforms, accurate tracking of the grid phase angle, and correct active/reactive current control. In addition, current tracking performance in the d-q frame reveals whether the controller responds quickly and remains stable when external conditions change.

Power quality is commonly evaluated using Total Harmonic Distortion. In grid-connected systems, low THD is essential because high harmonic content can negatively affect utility power quality, equipment operation, and compliance with interconnection requirements. For this project, THD is one of the main final evaluation criteria.

2.4 Standards and Grid Compliance Background

Inverter-based distributed generation systems are generally expected to satisfy accepted grid interconnection and power quality requirements. In this project, harmonic performance is evaluated against the commonly used project target of THD below 5%, which is consistent with the requirement adopted in the weekly progress evaluation.

Therefore, the literature and technical background reviewed for this project point to a clear need for an integrated solution that can combine effective MPPT, stable DC-link control, precise grid synchronization, strong current regulation, and low harmonic distortion. The proposed system architecture was developed in response to this need.

Chapter 3: METHODOLOGY AND RESEARCH PLAN

3.1 Methodology

The project was developed as a complete MATLAB/Simulink model of a three-phase grid-connected solar PV system. The main subsystems include the PV array, P&O-based MPPT controller, DC-DC boost converter, DC-link voltage control loop, three-phase inverter, Phase-Locked Loop, d-q current control block, and LCL output filter.

The PV array was first modeled to produce realistic voltage, current, and power characteristics under different operating conditions. The selected array configuration uses REC Solar REC215PE modules arranged as 11 series modules and 46 parallel strings. The I-V and P-V curves were observed to understand how the maximum power point shifts with temperature.

The MPPT controller was implemented using the Perturb and Observe algorithm. The algorithm continuously measures PV voltage and current, calculates PV power, and adjusts the converter duty ratio so that the operating point moves toward the maximum power point. The implemented logic also includes stable initial conditions and duty-ratio limits to avoid unstable operating behavior.

A boost converter was used to step up the PV array output and provide a suitable DC voltage for the inverter stage. The DC-link voltage was regulated around a 600 V reference using an outer PI controller. The error between the measured DC-link voltage and the reference value was processed through the controller to maintain a stable DC source under dynamic conditions.

For grid synchronization, a PLL was implemented to estimate the phase angle and frequency of the utility grid. Clarke and dq0 transformations were used so that the control system could operate in the rotating reference frame. Based on this synchronized angle, the current loop controller regulated d-axis and q-axis currents using PI control. The d-axis current was linked to active power transfer, while the q-axis reference was set close to zero to support unity power factor operation.

The controller outputs were transformed back to abc quantities and supplied to the PWM stage that drives the three-phase inverter. Finally, an LCL filter was placed between the inverter and the grid to reduce switching harmonics and improve the quality of the injected current. The completed model was then tested using dynamic irradiance profiles and the resulting waveforms were analyzed through scopes and FFT tools.

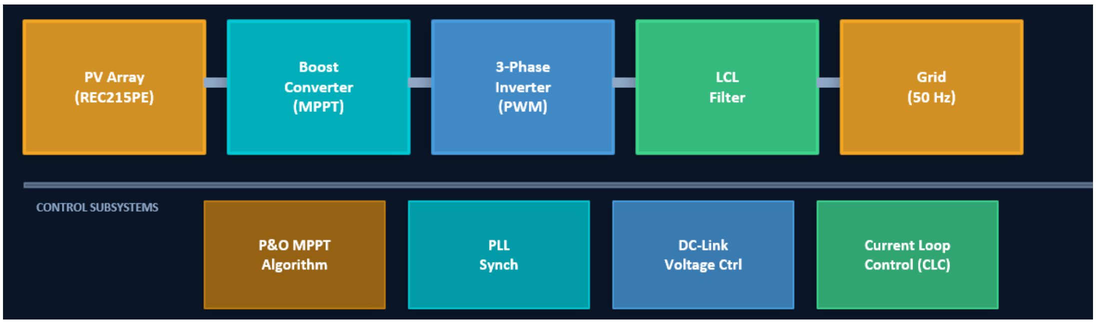

Figure 3.1. 1: Proposed system architecture

Table 3. 1: Key control and evaluation components of the proposed system

Stage	Function in the Proposed System
PV array	Generates DC electrical power from solar energy.
P&O MPPT	Tracks the maximum power point and produces the duty command for the boost converter.
Boost converter	Raises and conditions the PV output voltage.
DC-link controller	Maintains the DC bus around 600 V for stable inverter operation.
PLL	Synchronizes the inverter with grid voltage and frequency.
d-q current controller	Regulates active and reactive current components.
Three-phase inverter	Converts regulated DC power into AC for grid injection.
LCL filter	Attenuates harmonic components before power is delivered to the grid.

3.2 Timeline

The work progressed in stages from basic background study to full-system evaluation. The first stage included project selection, literature review, and identification of the main technical problem. The second stage focused on subsystem modeling, including the PV array, boost converter, and MPPT algorithm. The third stage involved inverter development, LCL filter integration, PLL implementation, and current loop control.

The final stage concentrated on performance evaluation, THD measurement, report writing, and preparation for final presentation.

Table 3. 2: Indicative timeline of major activities

Phase	Main Activities	Status
Phase 1	Project selection, literature review, research gap identification	Completed
Phase 2	PV array modeling, boost converter development, P&O MPPT implementation	Completed
Phase 3	Three-phase inverter modeling, LCL filter design, PLL and CLC integration	Completed
Phase 4	Dynamic testing, THD analysis, documentation and final evaluation preparation	Completed

	Phase (Week)	**6th Semester**														
	Activities	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
01	Project selection	■														
02	Literature review	■	■													
03	Identifying the research gap			■												
04	Modelling solar PV array & boost converter			■	■	■	■	■								
05	Implementation of P&O MPPT algorithm logic								■	■	■	■				
06	Three phase inverter modelling										■	■				
07	LCL filter designing										■	■	■	■		
08	PLL & CLC implementing												■	■	■	
09	DC- Link control												■	■	■	
09	Performance evaluation														■	
10	Documentation															■

Tasks completed

Future work

Chapter 4: PROGRESS TO DATE

4.1 System Development Progress

A complete MATLAB/Simulink model of the MPPT-controlled grid-connected solar PV system was successfully developed. The final integrated model includes the PV array, P&O-based MPPT block, boost converter, DC-link control stage, three-phase inverter, PLL subsystem, abc-to-dq and dq-to-abc transformations, current loop controller, and LCL output filter.

The project progressed from individual subsystem implementation to full integration and testing. Early work focused on MPPT and boost conversion, while later stages addressed PLL synchronization, d-q current regulation, and harmonic suppression. This staged development approach allowed each major function to be verified before the complete system was evaluated under dynamic operating conditions.

4.2 Detailed Technical Results

4.2.1 PV Array and MPPT Performance

The PV array characteristics were examined using I-V and P-V curves. At 25°C, the maximum power point was observed at a higher voltage than at 45°C, confirming that increasing temperature reduces open-circuit voltage and shifts the MPP to a lower operating voltage. This behavior clearly shows why dynamic MPPT is required in real PV systems.

A step irradiance profile was used to study system response under changing environmental conditions. The irradiance was reduced from 1000 W/m^2 to 500 W/m^2 at 0.2 s and then restored to 1000 W/m^2 at 0.3 s. During steady-state operation at 1000 W/m^2, the PV array delivered approximately 105 kW, with PV voltage close to 300 V and current near 350 A.

When irradiance dropped, the PV power reduced to approximately 50 kW, but the MPPT controller re-converged to the new maximum power point within about 50 ms. When irradiance returned to the original level, the controller quickly restored the operating point. These results show that the implemented P&O strategy was able to track the maximum power point effectively under dynamic conditions.

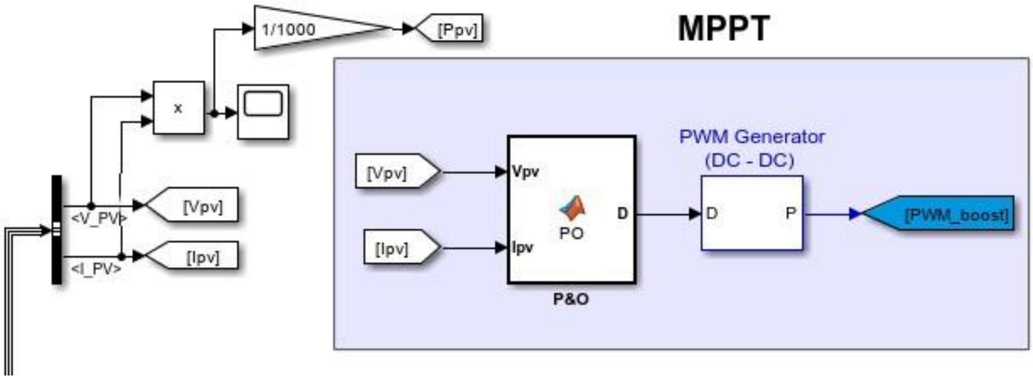

Figure 4.2.1. 1: MPPT Control Block (P&O Algorithm)

4.2.2 DC-Link Voltage Control and Grid Synchronization

The DC-link voltage controller was designed with a 600 V reference. Simulation results show that the DC-link reached its steady operating value after startup within about 50 ms. During the irradiance reduction at 0.2 s, the DC-link voltage dipped to approximately 480 V before recovering to the 600 V reference within about 30 ms.

At irradiance recovery, a brief overshoot to around 720 V was observed, but the system settled back to 600 V within approximately 25 ms. The steady-state measured value was 596.6 V, corresponding to a deviation of only about 0.57% from the reference value. This confirms that the outer PI voltage loop provided good regulation despite changes in available PV power.

Grid synchronization was achieved through the PLL subsystem. By using Clarke transformation and dq0 transformation, the controller was able to align the inverter output with the utility grid phase and frequency. This synchronization is essential for smooth real-power injection and for maintaining overall system stability.

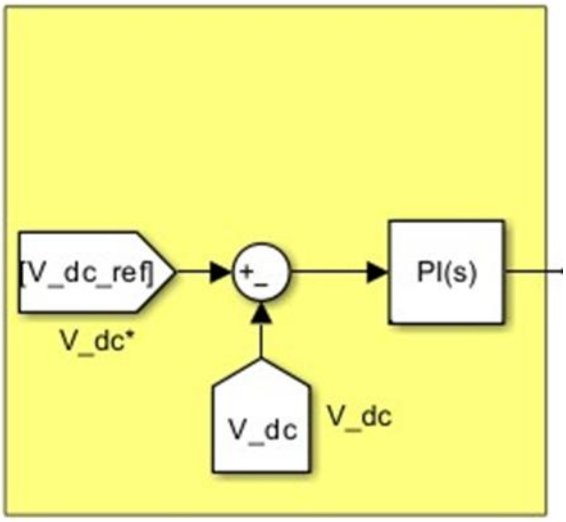

Figure 4.2.2.1: DC Link Control subsystem

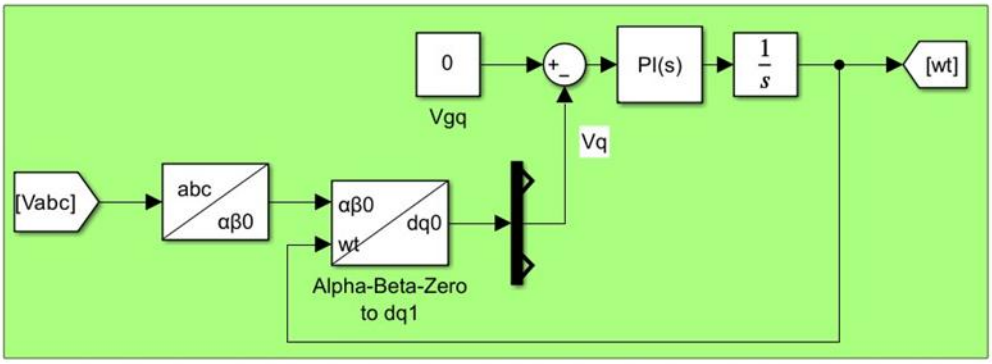

Figure 4.2.2.2: Phase-Locked Loop (PLL) Integration for Grid Synchronization

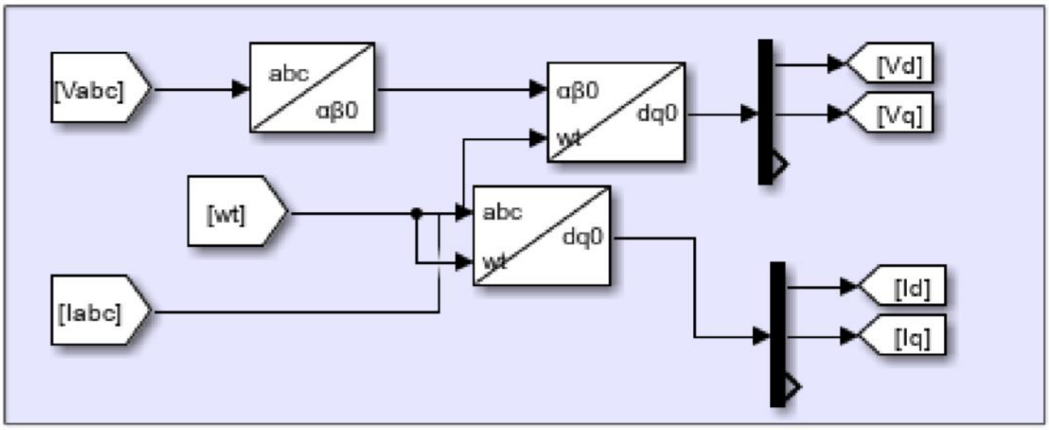

Figure 4.2.2.3: abc to d-q Transformation

4.2.3 Current Loop Control and Inverter Output Quality

The d-q current controller regulated the inverter current with good tracking performance. The d-axis current followed its reference closely, which indicates that the inverter was able to deliver the required active current to the grid. At the same time, the q-axis current was maintained close to zero, confirming unity power factor operation and negligible reactive power injection.

Following irradiance changes, the current loop settled within approximately 20 to 30 ms. This shows that the inner current loop responded quickly and helped stabilize the system after power disturbances. The generated PWM modulation signals were also consistent with the synchronized grid frequency.

At the grid side, the three-phase voltage and current waveforms were balanced and sinusoidal after the LCL filter. The voltage waveform was approximately 311 V peak, which corresponds to about 220 V RMS. The current waveform also reduced and recovered according to the available PV power, which further confirms correct converter and controller operation.

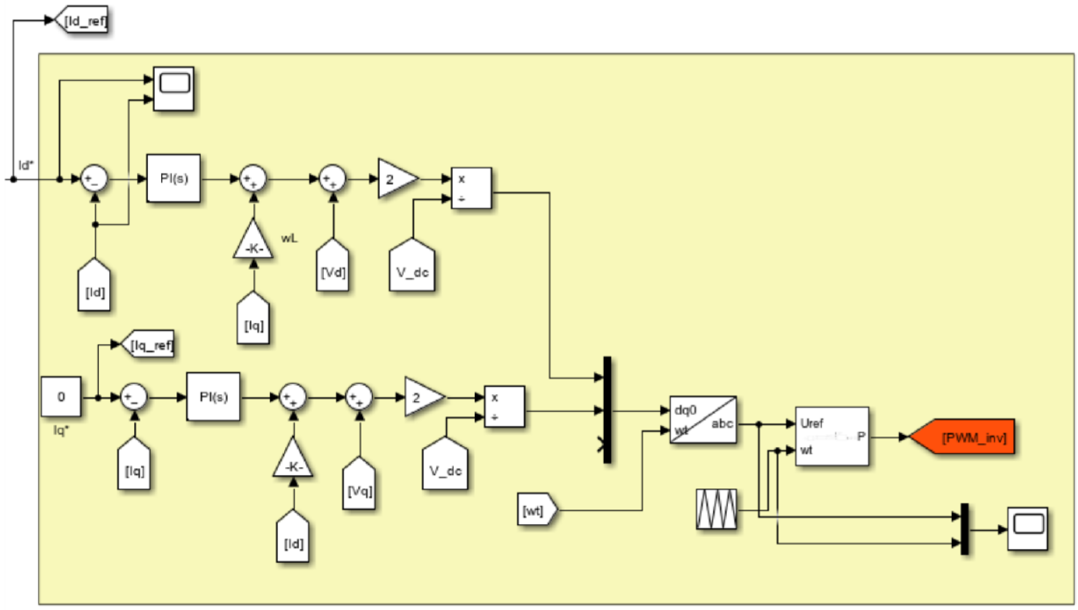

Figure 4.2.3. 1: Current Loop Control (CLC) System

4.3 Power Quality Analysis

Power quality was evaluated using FFT analysis of the steady-state grid current waveform. The analysis used the Simulink Powergui FFT Analyzer after the main transients had settled. The measured fundamental current was 109.1 A and the total harmonic distortion was 2.88%.

This THD value is well below the project target of 5%, indicating that the system satisfies the required harmonic performance level for the final evaluation. The harmonic spectrum also showed that the 5th harmonic had the highest relative magnitude at approximately 2.4%, while higher-order harmonics remained small.

The results therefore confirm that the LCL filter provided effective attenuation of inverter switching harmonics and that the overall control system produced clean and grid-compatible current waveforms. From a final evaluation point of view, this is one of the strongest outcomes of the project because it demonstrates both successful control integration and acceptable power quality.

Table 4. 1: Harmonic spectrum analysis and compliance results

Harmonic Order	Frequency (Hz)	Approx. Magnitude (% of fund.)	Result
3rd	150	~0.5%	PASS
5th	250	~2.4%	PASS
7th	350	~0.4%	PASS
11th	550	~0.9%	PASS
13th	600	~0.6%	PASS
17th	700	~0.7%	PASS
23rd	900	~0.1%	PASS
25th	1000	~0.1%	PASS

4.4 Simulink Model Overview

The complete MATLAB/Simulink model implemented for this project is shown below. It integrates all subsystems: PV Array, Boost Converter with MPPT (P&O), DC-Link Voltage Control, 3-Phase Inverter, LCL Filter, PLL, abc-to-dq transformation, and Current Loop Control.

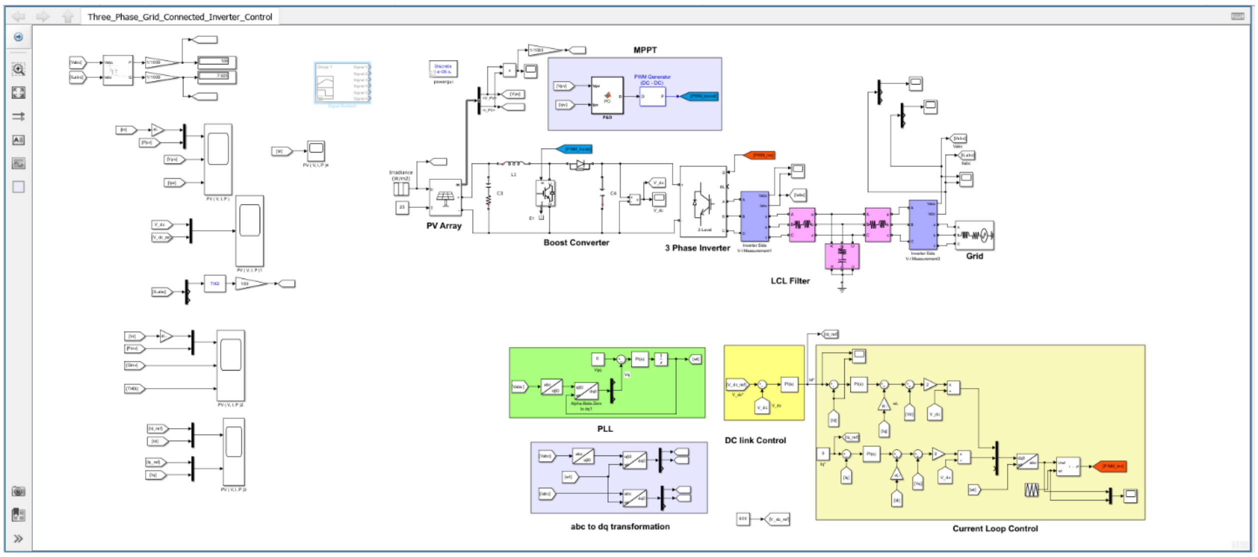

Figure 4.4. 1: Complete Simulink model – Three-Phase Grid-Connected Inverter with MPPT

4.5 Performance Evaluation

PV Array Characteristics (I-V & P-V Curves)

The PV array used in the simulation is the REC Solar REC215PE (BLK) module configured as 11 series modules and 46 parallel strings. The I-V and P-V characteristics at 1000 W/m² for two operating temperatures (25°C and 45°C) are shown below.

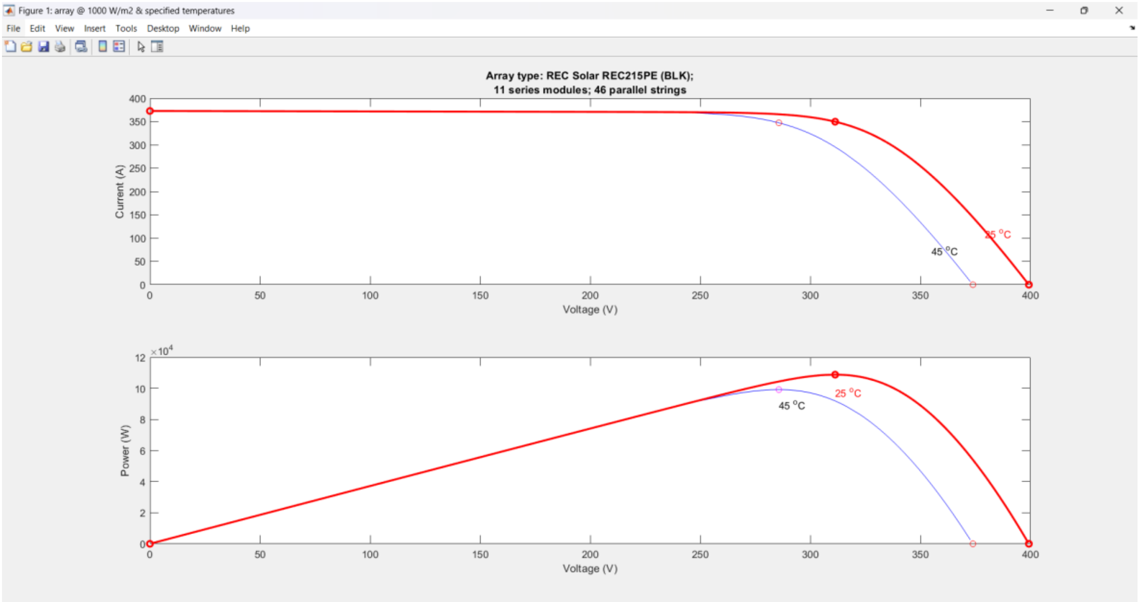

Figure 4.5. 1: PV Array I-V and P-V characteristics at 1000 W/m² – 25°C and 45°C .MPP shifts with temperature

- At 25°C, the array operates at a higher MPP voltage (~310 V) and peak power (~110 kW) compared to 45°C.
- Increasing temperature reduces open-circuit voltage (Voc) and shifts the MPP to a lower voltage, confirming the need for dynamic MPPT tracking.
- The P&O MPPT algorithm tracks these shifts continuously to extract maximum available power.

Irradiance Profile Applied

The irradiance profile applied during simulation consists of step changes to evaluate system response under dynamic conditions. The profile steps from 1000 W/m² down to 500 W/m² at t = 0.2 s, and recovers to 1000 W/m² at t = 0.3 s.

Figure 4.5. 2: Applied irradiance profile

PV Array Output – Voltage, Current, and Power

The PV array output voltage (V_PV), current (I_PV), and power are shown in response to the irradiance step profile. All three quantities track the irradiance changes, with the MPPT algorithm maintaining operation near the maximum power point.

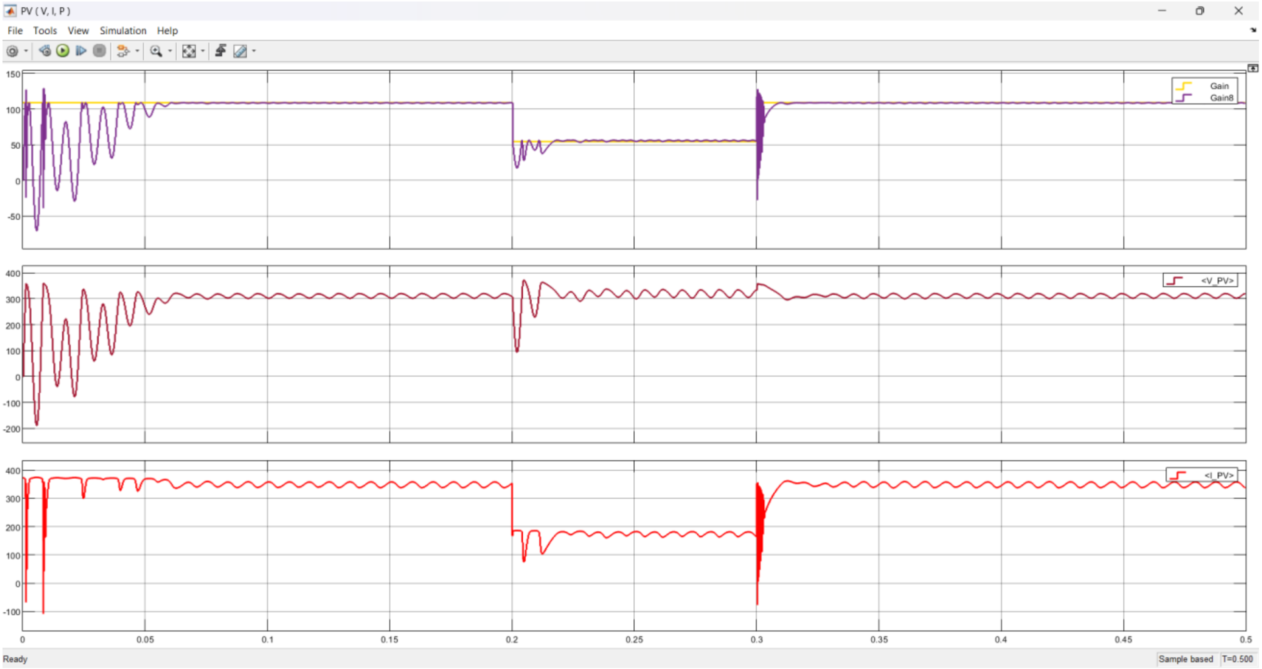

Figure 4.5. 3: PV array output: Power, Voltage V_PV, and Current I_PV under the irradiance step profile

DC-Link Voltage Regulation

Two scope outputs show the DC-link voltage (V_dc) regulation performance. The outer PI voltage control loop maintains V_dc at the reference of 600 V under the irradiance step changes.

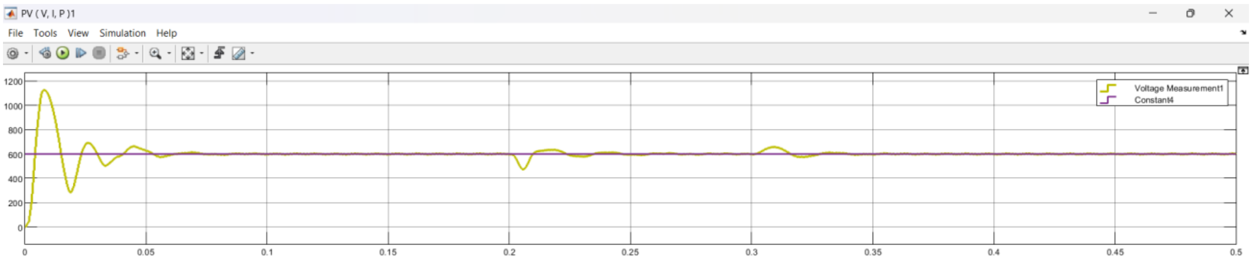

Figure 4.5. 4: DC-link voltage tracks reference closely under irradiance steps

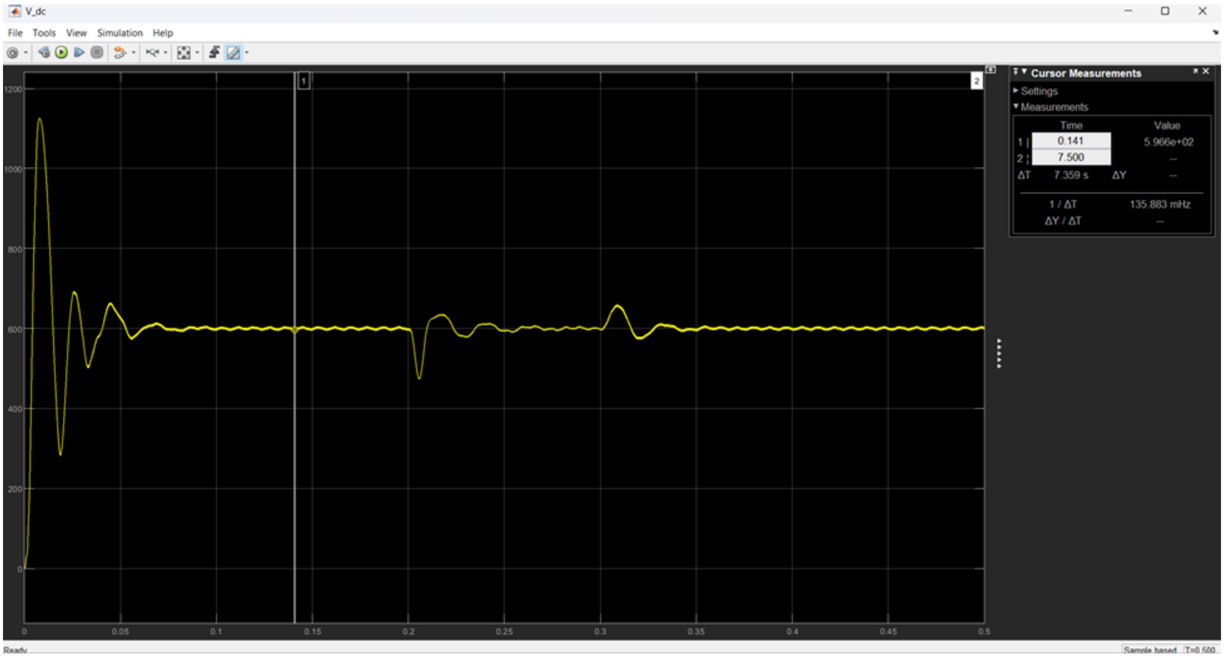

Figure 4.5. 5: V_dc waveform

- V_dc settles to 600 V after an initial transient within approximately 50 ms from startup.
- At the irradiance step-down (t = 0.2 s), V_dc dips to approximately 480 V before recovering to 600 V within ~30 ms.
- At the irradiance recovery (t = 0.3 s), a brief overshoot to ~720 V is observed, settling back to 600 V within ~25 ms.
- The cursor measurement confirms steady-state V_dc = 596.6 V, representing a deviation of only 0.57% from the 600 V reference.

d-axis and q-axis Current Tracking (CLC)

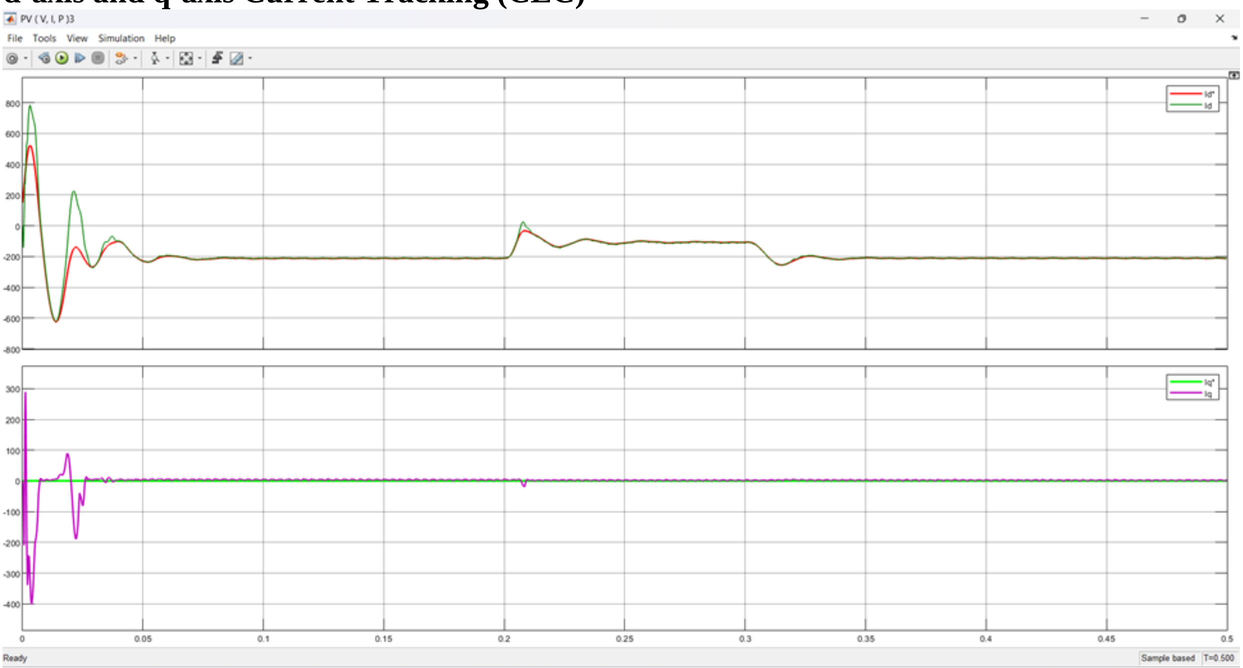

Figure 4.5. 6: dq current tracking: Id vs Id* and Iq vs Iq* — CLC maintains tight reference tracking

PWM Modulation Signal

Scope 4 confirms the PWM modulation signal generated by the current controller output, which drives the SVPWM block for the three-phase inverter.

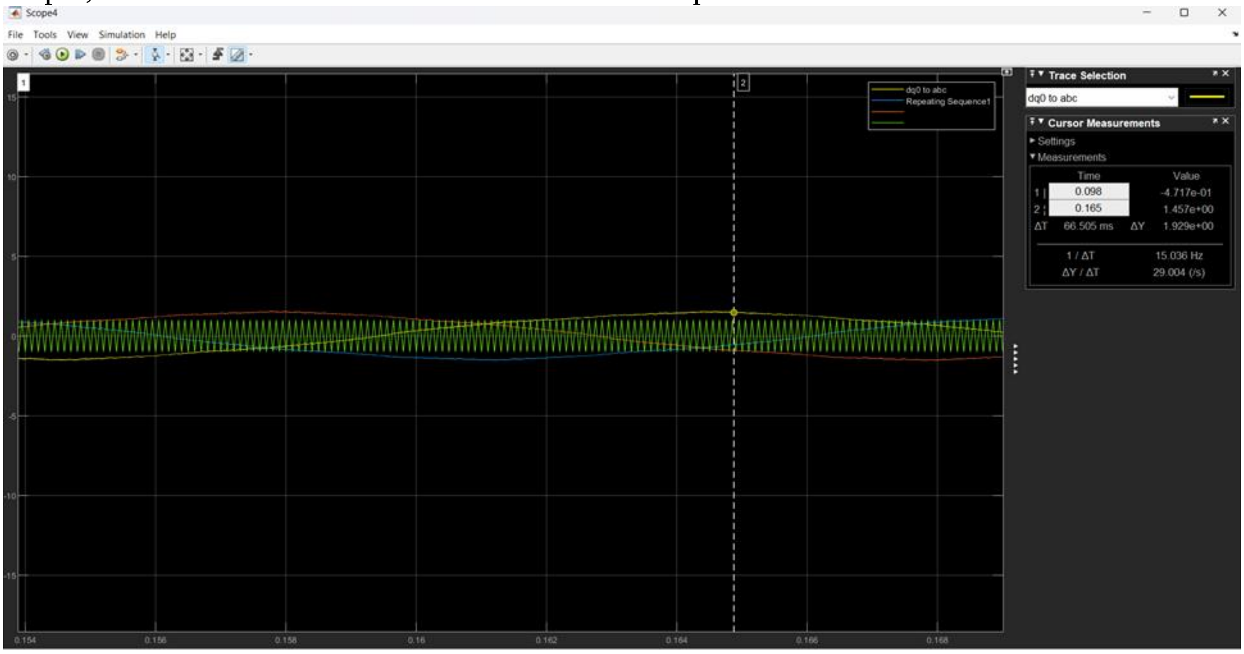

Figure 4.5. 7: Scope 4: dq0-to-abc transformation output and PWM repeating sequence — modulation signal for SVPWM

- The modulation signal closely follows the repeating sequence reference, confirming correct dq0-to-abc back-transformation.
- The signal amplitude and frequency are consistent with 50 Hz grid synchronization provided by the PLL.

Inverter Output – Three-Phase Voltage and Current

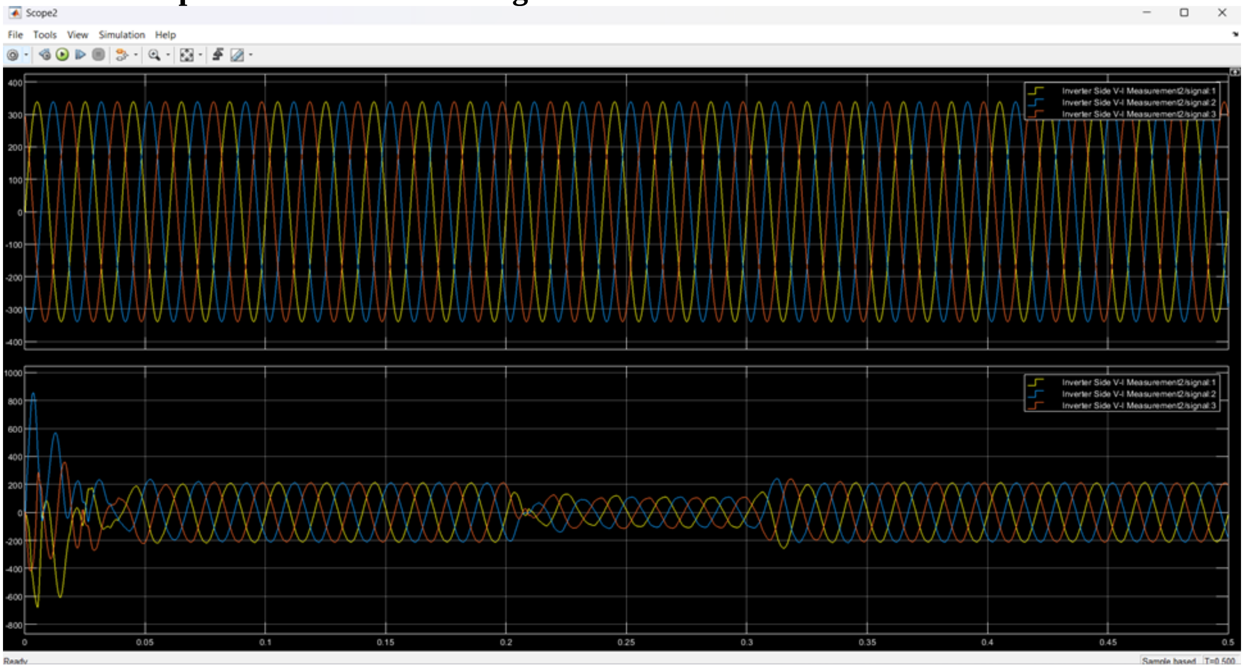

Figure 4.5. 8: Three-phase voltage and current at grid side – clean sinusoidal waveforms after LCL filter

- After the LCL filter (Scope 2), the voltage waveform is clean and sinusoidal with amplitude approximately 311 V peak (220 V RMS), matching the grid voltage.
- Grid currents are balanced three-phase sinusoids, confirming correct phase synchronisation by the PLL.
- Current amplitude reduces at the irradiance step (t = 0.2 s) and recovers at t = 0.3 s, consistent with PV power changes.

Chapter 5: ETHICAL CONSIDERATIONS

Although this project was carried out in a simulation environment, ethical considerations are still important because the work is related to real power system operation and renewable energy integration. In any future hardware implementation, electrical safety would be a primary responsibility. Proper protection, grounding, insulation, isolation, and safe grid interfacing must be ensured so that the system does not endanger users, equipment, or the utility network.

A second ethical aspect is power quality and grid responsibility. A grid-connected inverter should not inject poorly synchronized or highly distorted current into the utility system. For this reason, the project paid close attention to synchronization, current regulation, and THD reduction. The measured THD of 2.88% indicates that the simulated system remained within the selected harmonic target and therefore supports responsible grid interaction.

Environmental responsibility is another important consideration. Solar PV systems contribute to sustainable energy generation and can reduce dependence on fossil fuels. By improving the efficiency of power extraction through MPPT and maintaining good power quality during grid injection, the proposed system supports cleaner and more effective use of renewable energy. In a future practical version of the project, component selection, product lifetime, and responsible disposal of electronic materials should also be considered.

Finally, academic integrity is essential. All technical explanations, simulation work, and analysis presented in this report were prepared for educational purposes and based on the group's project development. External resources, software platforms, and supporting materials must be acknowledged honestly. Any use of AI-based writing assistance should be limited to language improvement, organization, and presentation support, while the actual engineering decisions and interpretation of results must remain the responsibility of the student group.

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